

10_Operators_Part1

```
# List of Arithmetic Operators in Python
# "+" for Addition
# "-" for subtraction
# "*" for multiplication
# "/" for division
# "%" for Modulus to get remainder of the division
# "**" for exponentiation
# "//" for floor division - rounded to nearest whole number
```

[2] ✓ 0.0s



```
a= 10
b= 3
print(a+b) # Addition
print(a-b) # Subtraction
print(a*b) # Multiplication
print(a/b) # Division
print(a%b) # Remainder of the division
print(a**b) # 2**2 = 4, 3**2 = 9
print(a//b) # 10/3=3.33 and floor value of 3.33 is 3
```

[4] ✓ 0.0s

```
... 13
    7
    30
    3.3333333333333335
    1
    1000
    3
```

```
# List of Comparison Operators in Python
# 1. == (equals to)
# 2. != (Not equals to)
# 3. < (Less than)
# 4. > (Greater than)
# 5. <= (Less than or equals to)
# 6. >= (Greater than or equals to)
```

[5] ✓ 0.0s

▷ ✓

```
x=5
y=10
print((x==y))
print((x!=y))
print((x<y))
print((x>y))
print((x<=y))
print((x>=y))
```

[6] ✓ 0.0s

```
... False
True
True
False
True
False
```

```
# Logical Operators in Python
# 1. and (Logical AND)
# 2. or (Logical OR)
# 3. not (Logical NOT)
```

[]

```
a=True
b=False
print(a and b)
print(a or b)
print(not b)
```

[8] ✓ 0.0s

```
... False
    True
    True
```

▷ ▾

```
x=5
y=3
print(x<10 or y > 10)
```

[9] ✓ 0.0s

```
... True
```

▷ ▾

```
# Assignment Operator
# = represents assignment and it assigns the value on the right to the variable on the left
# +=, -=, *=, /=, %= are compound assignment operators
a=10
b=3

c = a # simple assignment
b += 5 # Additional assignment and equivalent to b = b+5
a -= 2 # Subtraction assignment and equivalent to a = a-2
c *= 3 # Multiplication assignment and equivalent to c = c*3
b /= 2 # Division assignment and equivalent to b = b/2
a %= 4 # Modulus assignment and equivalent to a = a%4
c //= 2 # Floor Division assignment and equivalent to c = C//2
b **= 2 # Exponentiation assignment and equivalent to b = b**2
print("a = ",a)
print("b = ",b)
print("c = ",c)
```

[11] ✓ 0.0s

```
... a = 0
    b = 16.0
    c = 15
```

```

# Special Operators
# is and is not are the identity operators in python
# They are used to check whether two variables/values are located on the same part of the memory.

a = 5
b = 5
print( a is b)

list1 = [1,2,3,4,5]
list2 = [1,2,3,4,5]
print(list1 is list2)

name1="Hanu"
name2="Hanu"
print(name1 is name2)

```

[14] ✓ 0.0s

```

... True
False
True

```

▷ ▾

```

print(a,b)
print(id(a), id(b))
print(list1,list2)
print(id(list1), id(list2))
print(name1, name2)
print(id(name1), id(name2))

```

[16] ✓ 0.0s

```

... 5 5
10446536 10446536
[1, 2, 3, 4, 5] [1, 2, 3, 4, 5]
134863210210304 134863188760192
Hanu Hanu
134863188004512 134863188004512

```